

Brief Report: The Training and Assessment of Relational Precursors and Abilities (TARPA): A Preliminary Analysis

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Abstract The TARPA (Training & Assessment of Relational Precursors & Abilities) is a recently developed computer-based protocol for the assessment of a progression of key domains of responding critical to the development of generative language. In the current pilot study, five children with autism were assessed using the TARPA and their score on this protocol was correlated with ratings on the Vineland Adaptive Behavior Scales. Findings showed a statistically significant correlation. This and other features of TARPA performance and their implications for future testing and development of this protocol are discussed.

Keywords Assessment · Relations · Relational frame theory · Correlation · Vineland · Generativity

Understanding and accounting for the ability to produce and understand completely novel sentences—accounting for the generativity of language—is critical to any account of

language development (Malott 2003), and to the creation of programs for teaching flexible and fully functional language repertoires.

Relational Frame Theory (RFT; Hayes et al. 2001) explains linguistic generativity in terms of learned contextually controlled relational responding referred to as relational framing. Typically developing children learn relational framing through natural language interactions during which they are exposed to contingencies that establish these response patterns (e.g., Lipkens et al. 1993; Luciano et al. 2007). However, children with autism spectrum disorders (ASD) do not easily learn this key form of responding (e.g., Rehfeldt et al. 2007). Nonetheless, they can benefit from training of this repertoire (e.g., Murphy et al. 2005).

The ability to derive relations is implied in assessments of verbal behavior that look for novel responding and response generalization (e.g., Sundberg 2008), but direct testing of relational framing abilities is not yet typical in applied behavior analytic educational programs. The current research is a preliminary analysis of a protocol designed to enable direct assessment of this repertoire.

The current project attempted to evaluate the Training & Assessment of Relational Precursors & Abilities (TARPA) as a computer-based protocol for the assessment of a number of key forms of responding that are critical (from an RFT perspective) to the development of generative verbal behavior including (i) basic discrimination (ii) non-arbitrary [i.e., based on physical properties] conditional discrimination (iii) arbitrary [i.e., based on experimenter designated properties] conditional discrimination (iv) mutually entailed relational responding [e.g., deriving the symmetrical relation $B \rightarrow A$ from the trained relation $A \rightarrow B$]; (v) combinatorial entailed relational responding [e.g., derived the combinatorial relations $A \rightarrow C$ and

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$C \rightarrow A$ when trained with $A \rightarrow B$ and $B \rightarrow C$]; (vi) transfer¹ of function [responding to a stimulus in a new and appropriate way based on its participation in a derived sameness relation]. Each stage is further subdivided into multiple levels, and in the stages assessing derived relations (i.e., Stages 4, 5 and 6), levels are subdivided into training sections and derivation sections.

The aim of this study was to conduct a preliminary test of the TARPA with children with autism in order to correlate performance on this protocol with level of functioning as assessed using a standardized assessment tool (i.e., the Vineland Adaptive Behavior Scales) as well as to gain some insight into the hierarchical structuring and other features of the protocol to aid its further development and refinement.

Methods

Participants

Five² children (see Table 1 for ages and gender), each of whom had an independent diagnosis of autism, were recruited through a school for children with special needs at which the main researcher was employed. Consent for conducting the study was obtained from the school principal and parental consent was obtained for each participant. None of the children had taken part in similar research before.

Materials

Vineland II Adaptive Behavior Scales (VABS)

The VABS is a rating scale administered to caregivers to assess a child's daily living skills in three domains: communication (receptive, expressive and written), daily living skills (personal, domestic and community) and socialization (interpersonal relationships, play and leisure time, and coping skills). The VABS has excellent levels of split-half, inter-rater and test-retest reliability for each domain (Sparrow et al. 2005). A four point rating scale is used to score the response that best describes the behavior of the child that is being assessed on any given item (i.e., DK = do not know, 0 = never, 1 = sometimes and 2 = usually). An example of a VABS item (from the communication domain) is 'points to at least three major

body parts when asked (for example, nose, mouth, hands, feet, etc.).'

Training and Assessment of Relational Precursors and Abilities (TARPA)

The TARPA assessment was programmed in Visual BasicTM and was presented on a standard laptop. The protocol itself consisted of six stages.

Stage (1) tests the participant's ability to make simple discriminations (two visual and two auditory) and includes four sub-levels. An example of a task from this stage is the two comparison visual task (S1L1) in which a blank box and an abstract picture are presented simultaneously in random corners of the computer screen and the correct response is choosing the picture rather than the blank.

Stage (2) tests non-arbitrary conditional discriminations (i.e., in which the matches are physically similar) and includes two sub-levels (one auditory and one visual). An example of a task from this stage involving auditory stimuli (S2L2) is as follows: After a 'sample' button is clicked, a computer-generated nonsense word is heard and two further 'comparison' buttons appear successively in random corners of the screen, one accompanied by the same word as the sample and the other accompanied by a different word; the participant has to click the former.

Stage (3) tests arbitrary conditional discriminations (i.e., in which matches are not physically similar) and includes three levels (picture to word, word to picture and sound to word). An example of a task from this stage is the two comparison picture-to-word task (S3L1): After an abstract picture sample is clicked, two 'comparison' buttons appear successively in random corners of the screen, each accompanied by a different nonsense word, and the participant must click the button accompanied by the experimenter-designated correct word.

Stage (4) tests for mutually entailed relational responding and involves three levels (i.e., word to picture, picture to word and sound to word). As in Stages 2 and 3, a conditional discrimination format is used; however, each level in this stage involves both a training task and a testing task. An example of one of the levels is the two comparison sound-to-word level (S4L3): In the initial training task, the participant must pick a particular auditory nonsense word from an array (i.e., by selecting the correct button) when they hear a particular sound. In the subsequent testing task, they must show the derived 'mutually entailed' performance by selecting the sound from an array after hearing the nonsense word as sample.

Stage (5) trains and tests for combinatorial entailed relational responding and involves one level consisting of three sections (i.e., [i] word to picture training; [ii] sound to picture training; [iii] word to sound testing). The type of

¹ Transfer of function, which occurs in Stage 6 of the TARPA, is a special case of transformation of function in which the transformation is in accordance with relations of sameness or coordination. Hence, when referring to Stage 6, either 'transfer' or 'transformation' is technically correct.

² An additional participant refused to engage with the task and hence was dropped from the study.

Table 1 Participant sex, chronological age and scores on the VABS and TARPA

Pt. (sex)	Age	VABS scores				TARPA
		Communication	Daily skills	Socialization	Composite	
1(M)	13:6	25	48	34	33	1
2(M)	6:4	38	55	55	52	1
3(M)	7:7	77	100	23	66	7
4(M)	7:9	77	65	71	70	11
5(F)	10:2	100	63	68	75	12

conditional discriminative performance required in each section of this level is similar to those required in previous stages. In this stage however, the participant must derive the final (tested) conditional discriminative performance based on the two initially trained conditional discriminative performances. For example, in (i) the participant must choose a particular nonsense word (A) in the presence of a particular abstract picture (B); in (ii) he must choose a particular sound (C) in the presence of the same abstract picture (B) and in (iii) he must show ‘combinatorial entailment’ of the two taught performances to choose the word (A) with the sound (C) and vice versa.

Stage (6) tests for transformation of (mand) functions and involves one level consisting of three sections (i.e., [i] mand training; [ii] conditional discrimination training and derived testing; [iii] derived mand testing). The task employed was a computer-based adaptation of a procedure used by Murphy et al. (2005). A mand is a technical term which essentially denotes a ‘requesting’ response. An example of a performance seen in this stage is as follows. In (i) the participant learns to ‘mand’ using a particular token by clicking an onscreen button (in an array) that displays that token. From the current perspective, we would thus describe the token as having a particular ‘mand’ function for that participant. In (ii) the participant is trained and tested for the same kind of combinatorial entailed relation as seen in Stage 5 but this time the token is the A stimulus while a new novel arbitrary stimulus is the ‘C’ stimulus. Finally in (iii) the participant must mand using the ‘C’ stimulus (i.e., as previously done with the ‘A’), thus showing a transformation of the functions of the ‘C’ stimulus in accordance with the derived combinatorial entailed relation.

Procedure

The VABS was administered to the child’s caregiver by the experimenter. For TARPA assessment, the child and the experimenter sat in front of a laptop computer on which the TARPA was presented.

Each child required a number of assessment sessions. Session lengths were dependent on on-task behavior, the

length of time for which each child was available and progress on the TARPA. Specific information on number and approximate length of sessions is as follows: P1 [5 sessions: 15 min each]; P2 [11 sessions: 15 min each]; P3 [12 sessions: 15–20 min each]; P4 [3 sessions: 60 min each]; P5 [2 sessions: 90 min each].

P3, P4 and P5 used the mouse to select stimuli throughout the TARPA sessions. P1 and P2 had no mouse skills so therefore they simply pointed at on-screen stimuli and the experimenter used the mouse. Training trials included a continuous schedule of reinforcement for correct responding involving computer-delivered reinforcement and verbal praise. Reinforcement was provided throughout, except in the derivation sections of levels that assessed derived relations (i.e., particular levels of Stages 4, 5 and 6). Further to these forms of reinforcement, all participants were on individualized schedules involving the provision of breaks and edibles.

All TARPA levels involved a training element. For training, the experimenter first provided a demonstration trial in which she showed the participant how to respond correctly. Next, the experimenter facilitated a physically guided trial on which the participant was given assistance in responding correctly. On the third trial, the participant was given the opportunity to respond independently. If the participant performed correctly then they were exposed to further similar trials and if they reached a total of five consecutive correct then they were deemed to have passed that level. If a participant failed to perform correctly on any trial then they were re-exposed to the demonstration and guided trials before continued exposure to independent trials. If the participant got five cumulative incorrect responses then they were deemed to have failed that level. During trials assessing derived relations (which occurred in particular sections of levels throughout Stages 4, 5 and 6), there were no demonstration or guided trials (i.e., the participant received independent trials only).

For every TARPA stage to which they were exposed, participants were tested on all levels of that stage. If the participant failed every level, including every level section, of that stage then testing came to an end. If they passed any

level or level section of that stage then they were also exposed to the subsequent stage.

Results

Inter-observer agreement (IOA; a calculation based on the percentage of trials on which two independent observers agree as to the outcome of the trial) was calculated for all of the sessions in which the experimenter selected for the participant (i.e., sessions for P1 and P2). Trial by trial IOA was 98% for P1 and 99% for P2.

Table 1 shows each participant's score on the Vineland Adaptive Behavior Scales. Table 2 provides information (trials, etc) on each participant's performance on the TARPA. A Spearman's rank correlation test showed a strong and significant correlation ($r = 0.97$, $p = 0.05$) between performance on the VABS and the TARPA. Further tests were conducted to examine the relationship between TARPA performance and scoring for each of the sub-scales of the VABS. These showed that the TARPA correlated highly, though just outside significance ($r = 0.947$, $p = 0.056$) with the Communication sub-scale, and that it showed lower, more clearly non-significant correlation with the Daily living ($r = 0.56$, $p = 0.25$); and Socialization ($r = 0.56$; $p = 0.25$) sub-scales.

Discussion

The purpose of this study was to provide a preliminary test of the TARPA protocol using children with ASD as participants. With regard to the main correlation, future research involving a larger participant pool will be required to substantiate this finding; nevertheless, this was a good preliminary result for the TARPA. Future work will also need to utilize more robust tests in addition to the VABS (for example, the Pre-Language Scales 4 [PLS-4], Zimmermann et al. 2002) and to examine performances of typically developing control participants in addition to those of children with ASD in order to appropriately validate the TARPA as a protocol for the measurement of the relational skills underlying language. Nevertheless, even in the current context, the TARPA has performed in accordance with what might be expected of a language relevant protocol, since it correlated most highly (and closely approaching significance) with the Communication sub-scale of the VABS, while showing much lower correlations with the other sub-scales.

The current findings also provide some insight into the hierarchical ordering of the TARPA. Ideally, in any multi-stage hierarchically structured test, a participant should not be able to pass a stage of purported higher difficulty having failed all or most of the elements of a stage of purported lower difficulty. Inspection of the data of individual

Table 2 Participants' scores on the TARPA

Participant	Description	P1	P2	P3	P4	P5
Stage/level						
S1/L1	Simple dis. 2 visual comps	P (8) ^a	P (17)	P (8)	P (8)	P (8)
S1/L2	Simple dis. 4 visual comps	F	F	P (16)	P (13)	P (8)
S1/L3	Simple dis. 4 aud. (sound) comps	F	F	F	P (17)	P (8)
S1/L4	Simple dis. 4 aud. (word) comps	F	F	P (14)	P (8)	P (8)
S2/L1	Non. arb. cond. disc. 4 visual comps	F	F	P (24)	P (8)	P (8)
S2/L2	Non. arb. cond. disc. 4 aud. comps	F	F	F	P (15)	P (8)
S3/L1	Arb. cond. disc. 2 comps visual aud.			P (11)	P (17)	P (8)
S3/L2	Arb. cond. disc. 2 comps aud. visual			P (22)	P (8)	P (8)
S3/L3	Arb. cond. disc. 2 comps aud. aud.			F	P (8)	P (17)
S4/L1	Mut. entail. 2 comps derived aud vis			P (14)	P (8)	P (5)
S4/L2	Mut. entail. 2 comps derived vis aud			F	P (5)	P (5)
S4/L3	Mut. entail. 2 comps derived aud aud			F	F	F (5)
S5	Comb. entail 2 comps der. aud aud			F ^b	F ^b	F ^b
S6	Transfer of function via equivalence				F	P
Total		1	1	7	11	12

^a Numbers in brackets represent the number of trials required for the participant to pass that level

^b Though all three participants failed this stage, they each passed at least one sub-level (section), and thus, according to our criteria, all three were allowed exposure to the subsequent stage. Participants 4 and 5 did in fact progress to Stage 6, but participant 3 did not, due to experimenter oversight

participants across stages of the TARPA provided in the current report indicates that in general the TARPA conforms to this criterion. Once again, a larger participant pool would allow a more robust analysis of hierarchical structure. For example, with a substantially larger data set, order analysis (Krus et al. 1975) might be used to validate the hierarchy. Nevertheless, in the context of the limitations characterizing preliminary work, these results are still encouraging.

One feature of the current results which is both interesting in itself and also relevant as regards the future development of the TARPA is the apparent divergence of performance with respect to visual and auditory stimuli. A number of participants showed a pattern in which they appeared to show weaker responding with respect to levels involving exclusively auditory stimuli than levels involving at least some visual stimuli, irrespective of the strength of their overall performance. These results suggest that for at least some participants, pre-relational and relational responding with respect to auditory stimuli may be learned less readily than with respect to visual stimuli. This may be due to the fact that the tasks involving visual stimuli in the current protocol involved simultaneous discrimination whereas the tasks involving auditory stimuli involved successive discrimination. In any event this is an interesting pattern that is undoubtedly relevant with regard to both basic research and application.

In summary, this is a promising initial result for the TARPA and gives an early indication that this protocol could eventually provide a comprehensive tool for the assessment and training of core language skills in children with autism. The TARPA could thus constitute an important addition to existing programs for early intensive behavioral intervention for children with autism. Future

research will need to conduct a more comprehensive test of an evolved protocol using a larger sample size and controls. Such an analysis will also need to examine scores relative to more objective measures of intellectual functioning and language development.

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